Network Quality & Revenue Optimisation Analytics

Driving business growth through intelligent network performance analysis and strategic revenue enhancement

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Project Overview: Network Quality & Revenue Optimization

Introduction

Network quality drives customer satisfaction and revenue.

Aim

Analyze telecom network performance and usage data to identify revenue leakage and network congestion zones, then recommend optimization strategies using analytics and ML.

Goal

Identify key network factors affecting customer experience and revenue.

Our goal is to predict

high risk regions , custuomer usage behaviour , Predicting the ARPU , Complaint levels

Need

Proactive optimization to reduce churn and boost efficiency.

Predictive Analytics Report: Enhancing Network Performance and Revenue

Fetching Data from Data Lake

Extract raw network performance and customer usage data from centralized data lake.

Apply ETL to Apply Data Structure

Transform and clean data using Extract, Transform, Load processes for analysis readiness.

Store in Data Warehouse

Load structured data into optimized warehouse for efficient querying and analysis.

Apply ML Algorithm (Enhanced ML + XGBoost)

Deploy advanced machine learning models including XGBoost for predictive analytics.

WHAT WE ACHIEVED AND HOW WE IMPROVED OUR MODELS ACCURACY

Metric	Before Grid Search ($\text{Accuracy} = 83.33\%$)	After Grid Search ($\text{Accuracy} = 86.67\%$)	Change
Model	XGBoost Classifier	**XGBoost Classifier**	Same Model, Better Tuning
Method	Manual Hyperparameter	**GridSearchCV (Automated**	Optimized Parameter

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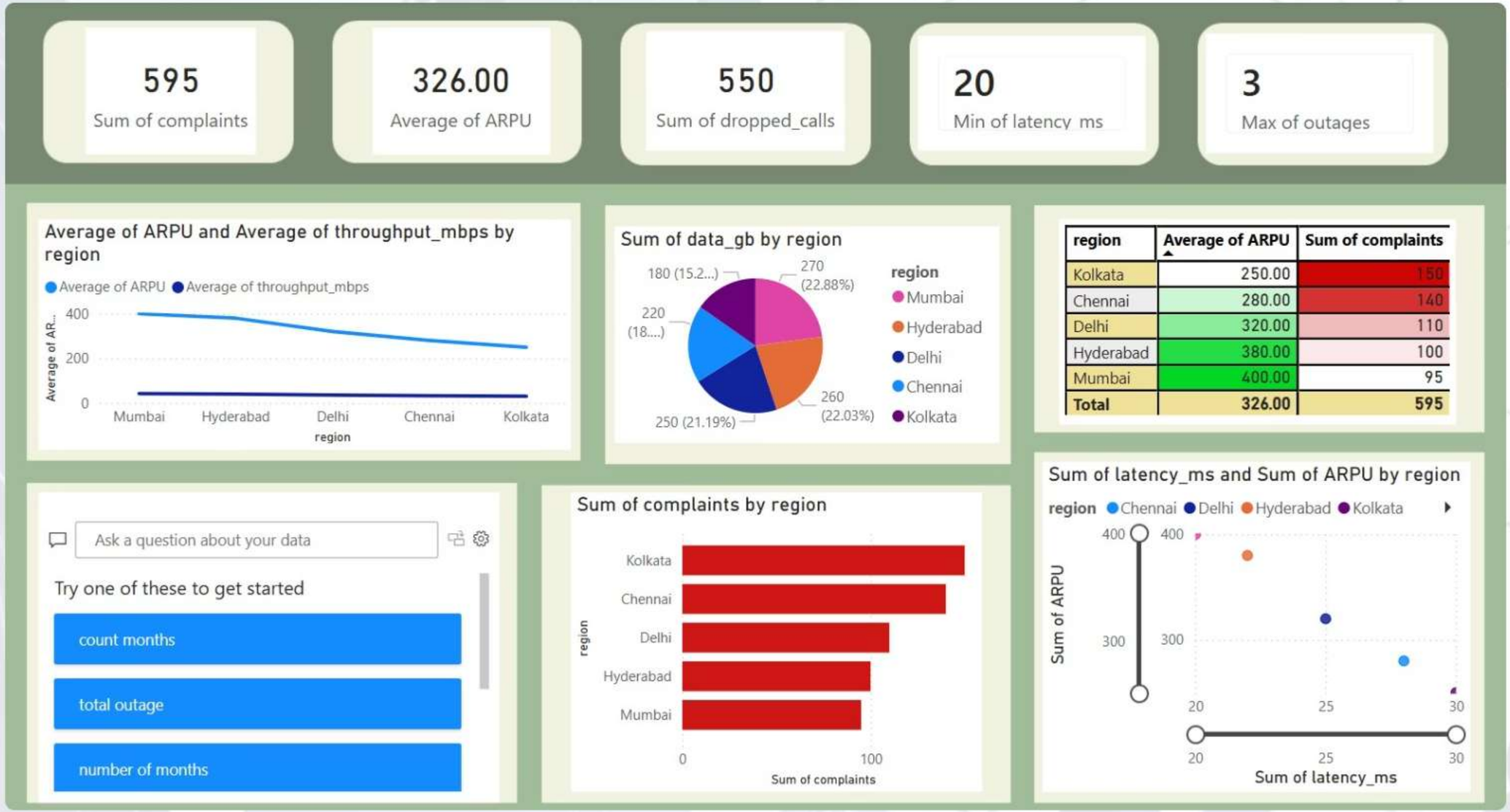
Model	XGBoost Classifier	**XGBoost Classifier**	Same Model, Better Tuning
Method	Manual Hyperparameter Selection	**GridSearchCV (Automated Tuning)**	Optimized Parameter Selection
Result	$\text{Accuracy} : 83.33\%$	$\text{Accuracy} : 86.67\%$	$+3.34\%$

```
✅ Best Parameters Found: {'learning_rate': 0.01, 'max_depth': 3, 'n_estimators': 500}

--- Optimized XGBoost Results ---
Accuracy: 0.8667

Confusion Matrix (Actual vs. Predicted):
[[14  1]
 [ 3 12]]

[Row 0: Actual Normal (0), Row 1: Actual At Risk (1)]
[Col 0: Predicted Normal (0), Col 1: Predicted At Risk (1)]

Classification Report:
              precision    recall  f1-score   support

           0       0.82      0.93      0.88        15
           1       0.92      0.80      0.86        15

    accuracy                           0.87        30
   macro avg       0.87      0.87      0.87        30
weighted avg       0.87      0.87      0.87        30
```

NETWORK AND REVENUE INSIGHTS



region	Average of ARPU	Sum of complaints
Kolkata	250.00	150
Chennai	280.00	140
Delhi	320.00	110
Hyderabad	380.00	100
Mumbai	400.00	95
Total	**326.00**	**595**

Our Recommendations and Improvements

Key Problems Identified:

- **Revenue Gap**: ARPU is ₹24 below target (₹326 vs ₹350 target) - 6.9% revenue shortfall
- **Customer Dissatisfaction**: 595 total complaints across network, indicating widespread service issues
- **Network Reliability Crisis**: 550 dropped calls causing direct customer frustration and churn risk
- **Regional Performance Disparity**:
  - Mumbai: Best performer (₹400 ARPU, 95 complaints)
  - Kolkata: Critical underperformer (₹250 ARPU, 150 complaints) - 60% revenue gap vs top performer
- **Infrastructure Instability**: Chennai shows 3 network outages (highest), directly impacting service quality
- **Usage-Revenue Disconnect**: Mumbai users consume 50% more data (270GB vs 180GB) but revenue optimization potential untapped

Strategic Solutions:

1

Critical Infrastructure Upgrades

Target **Kolkata & Chennai**. Prioritize latency reduction and deploy small cells to reduce dropped calls by 20% in Q4.

2

Revenue Optimization

In **Mumbai & Hyderabad**, launch premium tier upsells (e.g., Gaming Speed Boost, 4K Streaming Packs) leveraging stable networks.

3

Churn Prevention Program

Pilot a "Service Guarantee" in **Chennai**. Auto-credit 1GB data for customers experiencing >3 dropped calls/week to reduce churn.

4

Strategic KPI Shift

Shift focus to **complaint reduction** as the primary driver for improving ARPU, addressing the biggest current drag on revenue.